

Cybersecurity Basics

Modes of Learning

For this module, the following modes of learning will be used:

- Interactive Module
- Textbook Readings
- Online Quizzes

Introduction

Cybersecurity continues to be a concern across the regulatory spectrum. Organizations must understand this regulatory framework, to protect their sensitive data and preserve their legal, financial, operational and reputational interests.

Recent government initiatives in cybersecurity include:

Devoting more resources to creating and implementing a cyber response strategy.

New policy and advisory documents, emphasizing a collaborative approach between the public and private sectors.

New legislation relating to spam and disclosure of personal information, including mandatory breach notification requirements.

This module provides an overview of the legal landscape of cybersecurity in Canada, by explaining two main areas:

1. Industry statistics and Canadian cyberattacks.
2. Canada's cybercrime rhetoric and response to cybersecurity.

As you begin this module, please refer to the timeline and make note of any assessments or important dates. If you have any questions, ask your instructor.

Learning Outcomes

Upon completion of this module, you will be able to:

1. Describe the basic concepts and concerns common to cybersecurity.
2. Describe the different mechanisms and protection used in cybersecurity.

Key Terms & Concepts

Listed below are some important key terms and concepts within this module.

- Cybersecurity
- Cyberthreat
- Cybercrime
- Cyberattack

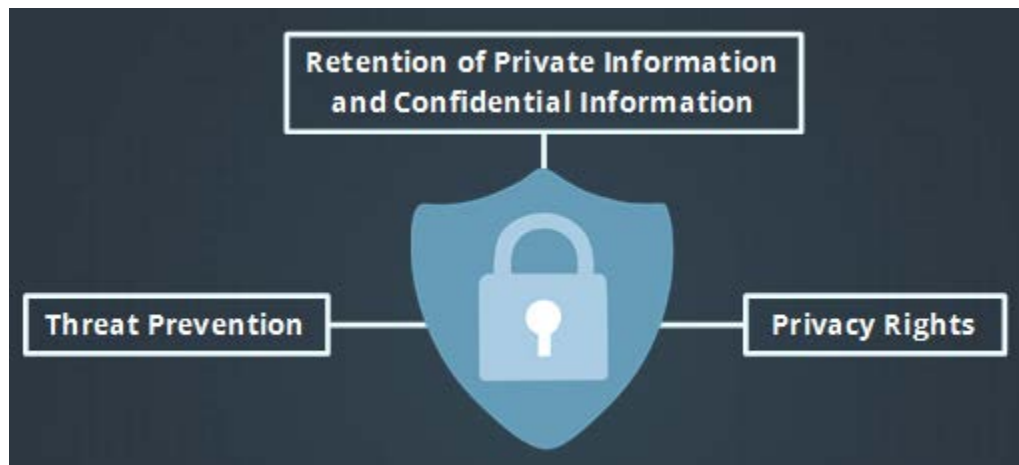
- Cyber Incident
- Data Breach

Meaning of Cybercrime

Cybercrime is broadly defined as a criminal offence that includes a computer as the main object of the crime or as the tool used to commit a material component of the offence (e.g. child pornography, hate crimes, computer fraud).

Cybersecurity Concepts and Concerns

Some of the major concerns in cybersecurity entail threat prevention, privacy rights and organizations' retention of private information. Organizations must understand the regulatory framework surrounding cybersecurity so they can protect their sensitive data and preserve their legal, financial, operational and reputational interests.



Information Protection Types and Mechanisms

Information Security originates from the top of an organization. The latter needs to have an Information Security Framework and Policies in place. The security/sensitivity of information needs to be classified; thus given the correct controls. The controls and protection mechanism are divided into broadly four categories: preventive, detective, corrective, and compensatory.

Industry Statistics and Canadian Cyberattacks

According to the Department of Finance, Canada has more computers per capita than any other country in the world, and Canadians are the heaviest internet users in the world, spending more than 40 hours online per month, per person.

This technological reliance increases the risk of cyberthreats to people and organizations in Canada. In fact, approximately 70% of Canadian businesses have been victims of cyberattacks, with an average cost of \$15,000 per incident.

The Global State of Information Security Survey 2016, conducted by PricewaterhouseCoopers, indicates that cybersecurity incidents in Canada have increased by 160%, year-over-year.

Furthermore, the president and CEO of the Canadian Council of Chief Executives has stated that the private sector is experiencing regular, steady attacks on cybersystems, occurring daily and repeatedly.

Several studies have noted cybersecurity breaches as a top risk to Canadian organizations. According to the Global Economic Crime Survey 2016, the Canadian Insights suggest that cybercrime was the second most common economic fraud perpetrated on Canadian organizations last year. In addition, the risk of cybercrime reportedly tops the list of frauds to occur in the next two years.

Some cyber professionals predict breaches that may result in extinction-level events, costing billions of dollars and irretrievably damaging reputations. These statistics and predictions all indicate that the growing threat of cybersecurity breaches requires organizations to be vigilant and informed.

Canada's Response to Cybercrime

In Canada, cybercrime is addressed by various partnerships between international, federal, provincial and municipal law enforcement agencies, including the Canadian Anti-Fraud Centre, the Ontario Provincial Police and the Competition Bureau of Canada (to name only a few). Together, these agencies combat internet fraud and mass-marketing fraud.

Public Safety Canada is a federal agency with the objective of protecting the economic prosperity, national security and quality of life in Canada.

Public Safety Canada is responsible for:

- Coordinating the implementation of the nationwide cybercrime strategy.
- Assessing emerging complex threats.
- Addressing risks within government and across Canada.

In response to growing cyber risks, Canada unveiled its cyber strategy in 2010, Canada's Cybersecurity Strategy: For a Stronger and More Prosperous Canada.

The strategy seeks to detect and prevent cyberattacks on public and private organizations and agencies across Canada.

It is based on three pillars:

1. Securing government systems.
2. Partnering to secure vital cybersystems outside the federal government.
3. Helping Canadians be secure online.

The Canadian Cyber Incident Response Centre (CCIRC) works with Public Safety Canada to advise on mitigating cyberthreats and to direct the national response to cybersecurity incidents.

The CCIRC collaborates with other federal agencies to identify risks and to determine potential interventions. It forms partnerships with provincial and territorial bodies and participants from private sectors, universities and non-governmental organizations.

In 2014, the federal government launched its Cybersecurity Cooperation Program (CSCP), a five-year, \$1.5-million initiative developed in support of Canada's Cybersecurity Strategy. The purpose of the

program is to improve the security of Canada's vital cybersystems by offering grants and contributions toward eligible projects to owners and operators, industrial and trade associations, academics, and research organizations.

Funding is provided to increase the availability of cybersecurity assessments, tools and methodologies for owners and operators of vital cybersystems. The fund from the program also aims to develop best practices for critical infrastructure sectors, and to support activities that contribute to improving the resilience of vital cybersystems outside the federal government.

Several government initiatives were launched in 2015 to strengthen cyber responses and to protect essential cybersystems outside the federal government, through enhanced public-private collaboration.

For example, the federal budget in the Economic Action Plan 2015 included two announcements on cybersecurity that both impact the private sector and devote significant resources to the issue of cyberattacks:

- i. The first announcement was \$58 million in new funding over the next five years, to further protect the Government of Canada's essential cybersystems and critical infrastructure against cyberattacks.
- ii. The second announcement was \$36.4 million in new funding over five years, to support the government's efforts to secure Canada's vital cybersystems. This funding will not go directly to the private sector but will be used to "provide enhanced support to operators through the development and dissemination of cybersecurity tools, security information and expertise."

The 2016 – 2017 Economic Action Plan also recognized the Canadian government's obligation to protect Canadians' personal and corporate information used to deliver public services.

The Plan proposes \$77.4 million over five years, starting in 2016–2017, to implement new measures to improve the security of government networks and information technology systems. These measures seek to ensure that the government can better defend its networks and systems from cyberthreats, malicious software and unauthorized access.

The federal government devised more initiatives in 2016 to combat cyberthreats. These initiatives aim to strengthen Canadian organizations' responses to cybersecurity threats, particularly in areas where Canada has been lagging behind other countries.

The Prime Minister's mandate letter to the Minister of Public Safety and Emergency Preparedness in December 2015 states that one of the Minister's top priorities is to "lead a review of existing measures to protect Canadians and our critical infrastructure from cyberthreats."

The federal government held a public consultation in 2016 to review measures to protect critical infrastructure and Canadians from cyberthreats.

Once the information is gathered, a report will be written to assist the government in developing cybersecurity policies and program decisions.

On May 6, 2016, Prime Minister Justin Trudeau endorsed the International Cybersecurity and Intelligence Conference organized by the Ontario College of Management and Technology. In his letter to the organizer, the Prime Minister acknowledged the need for cybersecurity professionals to come together to

learn, engage and discuss cybersecurity as one of the most prominent issues of our time. Over the past few years, the government has taken measures to translate these objectives into actions that included supporting the private sector.

One valuable initiative has been the development of advisory documents from the federal government such as:

- Get Cyber Safe Guide for Small and Medium Businesses
- Fundamentals of Cybersecurity for Canada's Critical Infrastructure Community
- Cyber Incident Management Framework for Canada

Get Cyber Safe Guide for Small and Medium Businesses

This document aims to “help Canadians who own or manage a small or medium business to understand the cybersecurity risks they face, and provide them with practical advice on how to better protect their business and employees from cybercrime.”

This document was developed by the National Cross Sector on Critical Infrastructure Working Group to recommend an appropriate response to the evolving landscape of cyberthreats. The document's guidelines were developed to address the need for action-oriented advice to increase cybersecurity. It is not a comprehensive document, and should be read in conjunction with other documents, such as industry-specific guidelines, international standards and other Government of Canada documents.

Fundamentals of Cybersecurity for Canada's Critical Infrastructure Community recommends that organizations take the following steps to strengthen their cybersecurity:

Identify Assets

- Which assets require enhanced or specialized safeguards?
- Maintain an inventory of all assets essential to operations.

Evaluate Threats and Risks

- What threats and risks could affect those assets or business operations?
- What safeguards should be in place to mitigate risks and secure assets?

Apply and Monitor Safeguards

Monitor safeguards and assets to prevent or manage security breaches.

Respond to Security Incidents

Respond to cybersecurity issues as they occur.

Make Adjustments When Needed

Update and adjust safeguards in response to changes in assets, risks and threats.

Get Cyber Safe Guide for Small and Medium Businesses

This document illustrates a collaborative approach that “sets out the roles and responsibilities of all levels of government, critical infrastructure owners and operations and other public and private sector partners, in the coordinate prevention and mitigation of, preparedness for, response to and recovery from incidents affecting Canada's portion of cyberspace.”

The private sector has also taken initiatives into devising collaborative efforts for responding and preventing cyberthreats. The Canadian Cyber Threat Exchange (CCTX) is a not-for-profit cybersecurity hub, which was launched in 2016;. Its founding members consist of Air Canada, Bell Canada, Canadian National Railway Company, Hydro One, Manulife, Royal Bank of Canada, TELUS, TD Bank Group and TransCanada Corporation.

CCTX's mandate is to respond collaboratively to cyberthreats, timely sharing information between Canadian businesses, government agencies and law enforcement, while also providing cyberthreat analysis and risk-mitigation recommendations. The goal of CCTX is to form a consolidated view of cyberthreats affecting Canada's private sector and act as a point of contact for cyber information-sharing organizations in other countries.

Module Summary

In this module you have learned:

1. Cybercrime is broadly defined as a criminal offence that includes a computer as the main object of the crime or as the tool used to commit a material component of the offence (e.g. child pornography, hate crimes, computer fraud).
2. Canada's rising reliance on internet technology increases the risk of cyberattacks to Canadian citizens and organizations. The current statistics and predictions indicate that the growing threat of cybersecurity breaches requires organizations to be vigilant and informed.
3. Progress in recent years by the Canadian government and private organizations has led to a holistic and cooperative strategy to mitigate the risk of cybersecurity breaches.

Knowledge Check

The following questions provide an opportunity for you to see what you remember and understand so far. Answer the questions to the best of your ability.

1. Cybercrime is a criminal offence that includes a computer as the main object of the crime, or as the tool used to commit a material component of the offence.
 - a. True
 - b. False
2. The major concern in cybersecurity is:
 - a. Threat prevention
 - b. Privacy rights
 - c. Retention of private information
 - d. All of the above
3. The three pillars of the national cybersecurity strategy are securing government systems; partnering to secure vital cyber systems outside the federal government; and helping Canadians be secure online.
 - a. True
 - b. False
4. The CCIRC works with Public Safety Canada to advise on mitigating cyberthreats and to direct the national response to cybersecurity incidents.

- a. True
 - b. False
5. The CCTX is not-for-profit cybersecurity hub that seeks to create a collaborative response to cyberthreats with the timely sharing of cyberthreat information between Canadian businesses and government agencies.
- a. True
 - b. False
6. Steps that organizations can take to strengthen their cybersecurity include:
- a. Identify Assets
 - b. Evaluate Threats and Risks
 - c. Applying and Monitoring Safeguards
 - d. Responding to Security Incidents
 - e. Making Adjustments When Needed
 - f. All of the above
7. A cyber incident could disrupt an organization's operations and damage its reputation. The organization might also face high remediation, investigation and litigation costs.
- a. True
 - b. False
8. Cyber incidents can be attributed to:
- a. Human error
 - b. Operational failure
 - c. Attacks by cybercriminals
 - d. All of the above

Answers:

- 1. The answer is True because cybercrime is a criminal offence which includes a computer as the main object of the crime, or as the tool used to commit a material component of the offence.
- 2. All of the above
- 3. The answer is True because the three pillars of the national cybersecurity strategy are security government systems; partnering to secure vital cyber systems outside the federal government; and helping Canadians be secure online.
- 4. The answer is True because CCIRC works with Public Safety Canada to advise on mitigating cyberthreats and to direct the national response to cybersecurity incidents.
- 5. The answer is True because CCTX is a not-for-profit cybersecurity hub which seeks to create a collaborative response to cyberthreats with the timely sharing of cyberthreat information between Canadian businesses and government agencies.
- 6. All of the above
- 7. The answer is True because a cyber incident may result in a disruption to business operations and damage to reputation. Additionally, cyber incident may leave the organization with high remediation, investigation and litigation costs.
- 8. All of the above

You have completed: Cybersecurity Basics

Remember to check the timeline before you proceed to the next module to ensure you have completed any assignments as required. Check with your instructor if you have any questions.